

Rodrigo Fernandez

Senior AI Data Engineer

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Summary

Senior AI Data Engineer with 12+ years of experience building cloud-native data platforms, ETL/ELT pipelines, analytics systems, and production-ready AI data workflows across SaaS, healthcare, eCommerce, and enterprise environments. Strong background in **Python, SQL, Apache Spark, Airflow, dbt, Kafka, AWS, GCP, Snowflake, BigQuery, PostgreSQL, and vector search**. Experienced in designing reliable data pipelines, model-ready datasets, RAG data flows, observability frameworks, privacy-aware data governance, and cost-efficient orchestration that turns raw operational data into trusted analytics and intelligent automation.

Skills

- **Programming & Data Engineering:** Python, SQL, PySpark, Bash, Pandas, NumPy, Pydantic, SQLAlchemy
- **AI / ML / LLM Data:** RAG pipelines, embeddings, vector search, feature engineering, model-ready datasets, OpenSearch Vector Search, pgvector, scikit-learn
- **Data Processing:** Apache Spark, Spark Structured Streaming, Kafka, Airflow, dbt, ETL/ELT, batch processing, streaming pipelines
- **Cloud & Warehousing:** AWS S3, Lambda, Glue, Redshift, CloudWatch, GCP BigQuery, Cloud Storage, Pub/Sub, Snowflake
- **Databases:** PostgreSQL, MySQL, SQL Server, Redis, DynamoDB, Elasticsearch/OpenSearch
- **DevOps & Quality:** Docker, Kubernetes, GitHub Actions, GitLab CI, Terraform, Great Expectations, OpenTelemetry, Prometheus, Grafana

Experience

Lightedge — Senior Software Engineer / AI Data Engineer *(Apr 2020 – Mar 2026)*

- Architected cloud-native **AI and analytics data pipelines** using **Python, Apache Spark, Airflow, dbt, Snowflake, and AWS S3** for SaaS, infrastructure, and finance-facing reporting workloads.
- Built production **LLM/RAG data workflows** by chunking operational documents, generating embeddings, storing vectors in **OpenSearch** and **pgvector**, and preparing trusted retrieval datasets for internal automation.
- Implemented real-time anomaly detection pipelines using **Kafka, Spark Structured Streaming, and Python scoring jobs**, reducing delayed incident discovery by **38%** across monitored customer data feeds.
- Refactored unstable **Airflow DAG factories** and oversized dynamic task generation, reducing scheduler failures and failed task retries by **46%**.
- Migrated legacy shell-based batch jobs into modular **Python packages, dbt models, and Airflow DAGs**, improving deployment control, maintainability, and recovery time.
- Designed privacy-aware ETL/ELT patterns with **column masking, schema contracts, Great Expectations checks, and IAM-scoped storage access** for sensitive customer datasets.
- Optimized Snowflake transformation workloads through incremental model tuning, clustering strategy, and warehouse sizing, reducing monthly compute cost by **29%** while maintaining SLA targets.

- Created reusable ingestion templates for APIs, SFTP drops, event streams, and warehouse extracts using **Python, Pydantic, Docker, and CI validation**.
- Fixed duplicate webhook processing that inflated revenue metrics by adding **idempotency keys, late-arrival handling, and reconciliation dashboards** across raw, staging, and curated layers.
- Delivered feature-engineering datasets for churn, billing-risk, and support-priority models using **Spark aggregations, dbt marts, and ML-ready warehouse tables**.
- Added pipeline observability with **OpenTelemetry, CloudWatch, Prometheus, Airflow alerts, and warehouse freshness checks**, improving failure visibility before reports became stale.
- Collaborated with product, security, DevOps, and analytics teams to convert unclear business requirements into governed data products and production AI workflows.

NIX United — Software Engineer *(Oct 2016 – Mar 2020)*

- Developed healthcare and eCommerce **data integration pipelines** with **Python 3.6–3.8, PostgreSQL 10–12, AWS S3, Redshift, Apache Spark 2.3–2.4, and Airflow 1.10** for multi-tenant client platforms.
- Implemented a patient appointment analytics pipeline that joined transactional records, timezone-normalized events, and notification logs, improving operational reporting accuracy by **34%** for scheduling teams.
- Fixed a critical **Spark 2.x memory error** caused by skewed joins on large order-history tables by salting high-cardinality keys, repartitioning datasets, and reducing job runtime by **41%**.
- Migrated cron-based CSV imports into **Airflow 1.10** DAGs with retries, task ownership, parameterized environments, and S3 landing zones, giving support teams clearer failure recovery paths.
- Built **Kafka 1.x–2.x** consumers for order, payment, and inventory events, then persisted validated payloads into PostgreSQL and Redshift models for near-real-time operational dashboards.
- Created backend-friendly data APIs with **Python Flask 1.x, SQLAlchemy**, and cached aggregate tables, helping React and PHP applications display reporting metrics without overloading transactional databases.
- Resolved duplicate customer-profile records by designing deterministic matching rules, checksum-based source tracking, and merge audit tables, improving downstream segmentation consistency by **27%**.
- Implemented incremental warehouse loading with watermarks, soft-delete detection, and SCD Type 2 dimensions, allowing analysts to compare historical account states without rebuilding full datasets.
- Improved data security by separating raw and curated buckets, tightening IAM roles, encrypting S3 objects, and removing hardcoded credentials from older deployment scripts.
- Supported model-ready datasets for fraud and product recommendation experiments using **scikit-learn 0.20–0.22**, pandas feature preparation, and SQL-based validation before handoff to data science teams.
- Partnered with QA and business analysts to validate edge cases in billing, appointments, and inventory reports, turning ambiguous defects into reproducible data fixes and regression checks.

Webiz Digital — Software Developer *(Jan 2014 – Sep 2016)*

- Built early data engineering workflows for digital commerce and marketing analytics using **Python, pandas, MySQL, PostgreSQL, and Linux cron jobs**.
- Created SQL reporting datasets combining orders, campaign clicks, customer profiles, and payment statuses to reduce manual spreadsheet reporting.
- Fixed malformed CSV import failures by adding schema validation, encoding normalization, and rejected-row logging for support investigation.
- Migrated manual Excel cleanup into scripted **Python transformations, reusable SQL views, and scheduled exports**, reducing weekly analyst preparation effort by **31%**.
- Designed normalized MySQL tables and aggregate reporting views for product, customer, and order metrics, improving dashboard query performance.

- Built predictive scoring prototypes with **scikit-learn, logistic regression, and cleaned campaign data** to support lead-prioritization experiments.
- Improved slow reporting queries by adding composite indexes, rewriting subqueries into joins, and archiving inactive records, improving dashboard response time by **36%**.
- Added reconciliation checks between payment exports and internal order tables, catching missing settlement rows before finance reporting cycles.
- Documented ingestion steps, rollback notes, and SQL validation queries so developers could support recurring data issues more easily.

Aurio — Junior Software Developer (*Oct 2012 – Dec 2013*)

- Supported internal reporting systems using **Python, SQL Server, MySQL, stored procedures, and Linux automation** for operations and customer-support teams.
- Cleaned customer, ticket, and transaction records using SQL scripts and Python utilities for monthly management reporting.
- Fixed recurring report mismatches caused by inconsistent timezone conversion by standardizing UTC storage rules and documenting downstream logic.
- Created stored procedures and indexed views for high-use operational reports, reducing repeated manual joins.
- Assisted migration of spreadsheet-based tracking into relational database tables with validation rules, import scripts, and review queries.
- Built automation scripts for log parsing, file movement, and scheduled report generation, improving daily data availability.
- Investigated duplicate ticket-count issues by tracing joins across customer and activity tables, then adding grouping logic and QA checks.
- Learned production support discipline by checking failed jobs, reading logs, reproducing data defects, and escalating fixes with clear evidence.

Microsoft — Software Developer Intern (*Jun 2012 – Sep 2012*)

- Supported internal software tooling with **Python scripting, SQL queries, API testing, and backend debugging**.
- Assisted senior engineers with data validation scripts and automation tasks that improved repeatable checks across internal development workflows.
- Gained experience with enterprise engineering practices, source control, documentation, testing discipline, and structured code review.
- Reproduced and helped resolve early-stage bugs related to form validation, inconsistent SQL query results, and input-handling logic.

Education

Texas State University

Bachelor's Degree in Computer Science (*Sep 2008 – May 2012*)